

Machine Learning & Deep Learning Project

Objective

1. Choose a **tabular dataset** (≥ 3000 rows)
2. Build:
 - **Machine Learning models**
 - **Deep Learning model (ANN)**
3. **Compare both approaches**
4. Justify:
 - When ML is better
 - When DL is better

All datasets are **tabular** and suitable for **classification or regression**.

☒ Option 1: Credit Risk Prediction (Classification)

- Dataset: Credit Card / Loan Data
 - Task: Predict if a customer will **default (0/1)**
 - Features: income, age, loan amount, credit history
 - Real-world value: Banking risk systems
-

☒ Option 2: House Price Prediction (Regression)

- Dataset: Housing prices dataset
 - Task: Predict **house price**
 - Features: area, location, rooms, year built
 - Real-world value: Real estate analytics
-

☒ Option 3: Customer Churn Prediction (Classification)

- Dataset: Telecom / subscription data
- Task: Predict if customer will **leave (churn)**
- Features: usage, tenure, plan type
- Real-world value: Retention strategies

☒ Option 4: Medical Insurance Cost Prediction (Regression)

- Dataset: Insurance dataset
 - Task: Predict **insurance charges**
 - Features: age, BMI, smoker, region
 - Real-world value: Pricing systems
-

☒ Option 5: Employee Attrition Prediction (Classification)

- Dataset: HR dataset
 - Task: Predict if employee will **leave**
 - Features: salary, job role, satisfaction
 - Real-world value: HR analytics
-

☒ Option 6: Sales Forecasting (Regression)

- Dataset: Retail / store sales
 - Task: Predict **future sales**
 - Features: date, promotions, store type
 - Real-world value: Business planning
-

◇ Part 2: AI Project Pipeline (AI Plan)

Students must follow this **strict pipeline**:

1. Problem Definition

- Define:
 - Problem type (classification / regression)
 - Input features
 - Output variable
 - Success metric (Accuracy, F1, RMSE)

2. Data Reading

3. Data Cleaning

Tasks:

- Handle missing values
- Remove duplicates
- Fix data types

4. Exploratory Data Analysis (EDA)

Required:

- Distribution plots
- Correlation heatmap
- Target analysis

5. Data Preprocessing

- Encoding categorical features
- Feature scaling

6. Train/Test Split

7. Model Building

Students must try **at least 3 models**

Classification:

- Logistic Regression
- Random Forest
- XGBoost

Regression:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting

8. Model Evaluation

Classification:

Regression:

9. Model Optimization (IMPORTANT)

- Hyperparameter tuning (GridSearch / RandomSearch)
- Feature selection

10. Final Model Selection

- Compare models
- Choose best based on metric

◇ Part 3: Code Organization (MANDATORY)

Students must structure their project like this:

```
project/
├── data/
│   └── dataset.csv
├── notebooks/
│   └── analysis.ipynb
├── src/
│   ├── data_cleaning.py
│   ├── preprocessing.py
│   ├── model.py
│   └── evaluation.py
├── README.md
├── requirements.txt
└── report.md
```

◇ Part 4: Markdown Documentation (README.md)

Students must include:

Project Title

Introduction

Describe the problem

Problem Statement

What are we solving?

Dataset Description
Explain features

Methodology
- Data Cleaning
- EDA
- Modeling

Results
Metrics and comparison

Conclusion
Insights and future work

◇ Part 5: Presentation Slides Structure

Students must prepare slides with:

Slide 1: Title

- Project name
 - Student names
-

Slide 2: Introduction

- Background
 - Why this problem matters
-

Slide 3: Problem Statement

- Define ML task
-

Slide 4: Dataset

- Features
 - Size (>3000 rows)
-

Slide 5: Data Cleaning & EDA

- Key insights
 - Visualizations
-

Slide 6: Methodology

- Models used
 - Pipeline
-

Slide 7: Results

- Metrics comparison
-

Slide 8: Proposed Solution

- Best model
 - Why chosen
-

Slide 9: Evaluation

- Strengths & weaknesses
-

Slide 10: Conclusion

- Final insights
 - Future improvements
-

◇ Extra Constraints (Important)

- Dataset must be ≥ 3000 rows
- Must use **at least 3 ML models**
- Must include **visualizations**
- Must achieve **reasonable performance**
- Code must be **clean and documented**